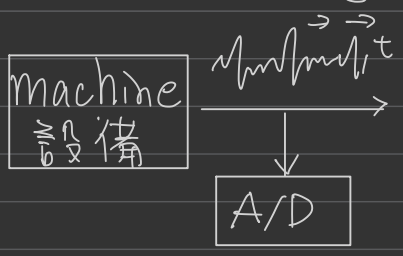
Example:

振動 signal



Task: OK or NG?

OK:

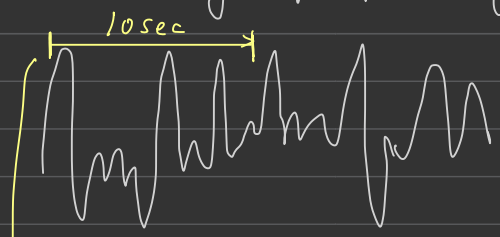
NG:

Step 1. Data engineering

digital signal

sensing, preprocessing

公司最珍貴 raw data



segmentation (window)

preprocessing
filtering ... etc

preprocessed signal

$X[t]$, $t=0, \dots, N-1$

$X[t] \longrightarrow$ 特徵提取 featured extraction $\longrightarrow$ ① SD值(標準差)

$\longrightarrow$ ② μ值

$\Downarrow$

10sec signal

if $f_s = 1\,kHz$

$\Downarrow$

10k sample points $\in \mathbb{R}^{10000}$ 維度縮減 (dimension reduction)

$\longrightarrow \vec{X} = \begin{bmatrix} SD \\ \mu \end{bmatrix} \in \mathbb{R}^2$ ... 一筆 data / 一個 feature vector / 一筆 data point

特徵提取

$X_1[t] \rightarrow \vec{X}_1$

$X_2[t] \rightarrow \vec{X}_2$

$\boxed{\text{Training}}$ step 1: collect data

$\vec{x}_1, \dots, \vec{x}_{10}$ 在 $\boxed{\text{OK下}}$ 收集 → 一個類別 class → negative class ⇒ −1

data set

label

$\vec{x}_{11}, \dots, \vec{x}_{20}$ 在 $\boxed{\text{NG下}}$ 收集 → 另一個類別 → positive class (陽性) ⇒ +1

→ 2個類別

$$\boxed{S} = \{(\vec{x}_i, \vec{y}_i)\}_{i=1}^{20}, \quad \vec{x}_i \in \mathbb{R}^2$$

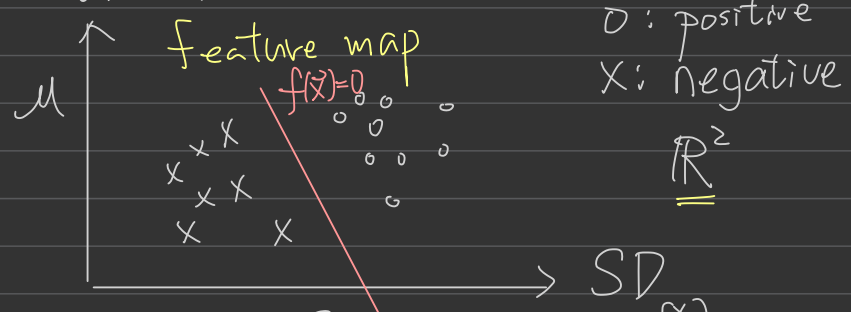
where $\vec{x}_i$: training data (points)

$\vec{y}_i$: class label for i training data

→ Training Set

Step 2: Modeling 建模

observation,



o : positive

x : negative

$\mathbb{R}^2$

散佈圖 ⇒ 特徵視覺化

scatter plot 想用肉眼觀察特徵好或不好

Assume : $f(\mu, \sigma)$ → 假設一個 model

$\Rightarrow \mu = a\sigma + b$

有無限多种 (a, b) 答案

模型參數

or 超參數

hyper parameter

Q：怎麼找 (a^*, b^*) 參數？

$$\underbrace{S \longrightarrow \boxed{ML} \longrightarrow (a^*, b^*)}_{\text{Model training (data-driven)}}$$

$$\mu = a^* \sigma + b^*$$

$$\Rightarrow \mu - a^* \sigma - b^* = 0$$

$$\Rightarrow \underbrace{[1 \quad -a^*]}_{\triangleq \vec{w}^T} \underbrace{\begin{bmatrix} \mu \\ \sigma \end{bmatrix}}_{\vec{x}} + (-b^*) = 0$$

$$\Rightarrow f(\vec{x}) = \vec{w}^T \vec{x} + b = 0$$

→ 超平面 (hyperplane)

⇓

決測邊界 (decision boundary)

⇒ 找到 a^*, b^* → end of training

Testing：

10 sec

↓

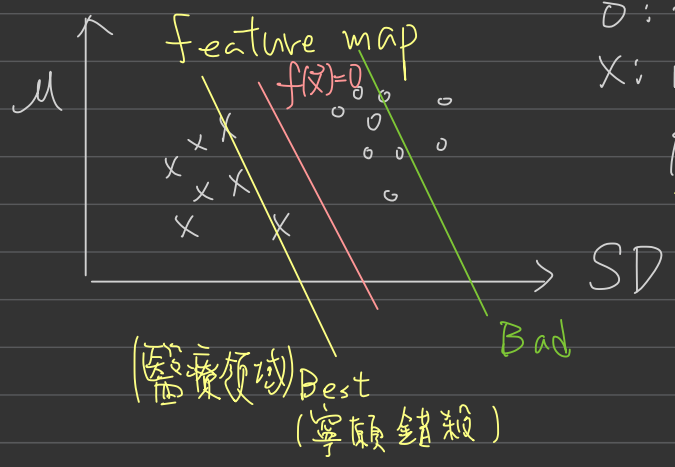
feature extraction

$\vec{x} \in \mathbb{R}^2$ ... label is unknown

classification 分類 ⇔ classifier

$\boxed{f(\vec{x}) : \text{分類器}}$ ⟶ $\begin{cases} f(x) > 0, & \vec{x} \text{ 被分類成正類別} \\ f(x) < 0, & \vec{x} \text{ 被分類成負類別} \end{cases}$

Q: model 的性能 (performance)?



o: positive
x: negative

$\mathbb{R}^2$

Ans: domain-dependent

Model Generalization 模型泛化能力

⇒ performance on real data